Security Noir: Critical Designs for Security & Privacy

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In this paper, we investigate an alternative method for communicating the somewhat elusive security risks and privacy invasions of technology, the web, social networks, and other aspects of modern life. Specifically, we draw from the *critical design* movement popularized by Dunne and Raby in the 1990s. In this project, we have conceptualized 10 new technological designs and described their use through short vignettes. These designs are imaginary but plausible technologies that illustrate a security or privacy risk through absurdity, satire, or humour.

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1 Introductory Remarks

Consider the following vignette from a collection of speculative artefacts:

> **Eleventh Finger.** Biometric authentication based on fingerprints is generally user-friendly and fast. Eleventh finger is a 3D printed rubber finger, customized with the user's fingerprint. You can put it on a keychain and use it on cold winter days when you do not want to remove your gloves. Alice lets her friend Bob borrow it when he stays at her house. It can also serve as a backup if the worst happens.

At face value, the design is a convenience-enabling accessory for biometric authentication; yet, the vignette also challenges common assumptions about fingerprints as secure, embodied identifiers. What happens when bodily identity is lent, misplaced, duplicated, or stolen? The scenario does not present a technical solution; instead, it invites reflection.

This mode of engagement is characteristic of critical design, which involves speculative artifacts and narratives that do not propose deployable products, but rather provoke reflection and discussion. Early work on critical design is commonly attributed to Dunne and Raby [2, 4], and it can also be seen as a refinement of earlier movements in arts and design, beginning with the Italian radical design movement and extending through critical practice in HCI [11].

Designing security software and privacy-enhancing technologies requires anticipating threats that have not yet been exploited, a process in computer security known as *threat modelling* [15][CITE Shostack, 2014]. While valuable, this work is often time-consuming for security experts and difficult for practitioners to learn [16] [CITE Shostack, 2021]. Prior work has therefore explored bridging speculative design with threat identification, highlighting that threat modelling is inherently speculative, requiring the imagination of possible futures [12] [CITE Merrill et al., 2020]. From this perspective, tools from speculative design and design fiction can support early-stage and exploratory threat identification.

These approaches are also more lightweight than traditional frameworks, making them accessible to practitioners with limited security expertise. This aligns with calls within industry and practice for "fast, cheap, and good" approaches to threat modelling. For example, Shostack + Associates (2021) advocate starting with a simple but generative question:

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Table 1. An Example of Critical Design

Work	Vignette
Compass Table, A. Dunne F. Raby	This table reminds you that electronic objects extend beyond their visible limits. The 25 compasses set into its surface twitch and spin when objects like mobile phones or laptop computers are placed on it. The twitching needles can be interpreted as being either sinister or charming, depending on the viewer's state of mind. When we designed the compass table, we wondered if a neat-freak might try to make all the needles line up, ignoring the architectural space of the room in favour of the Earth's magnetic field." Quoted from [4].

"What can go wrong?" Speculative methods operationalize this question through narrative and imagination, helping make abstract or overlooked risks more visible.

In this work-in-progress, we investigate critical design as a lightweight and narrative-driven approach to communicating elusive security and privacy risks. This work makes two primary contributions. First, we present a curated corpus of ten speculative design vignettes that surface a range of security and privacy tensions in everyday life. Each artefact is constructed to make a particular socio-technical risk perceptible through exaggeration, satire, or narrative juxtaposition, enabling readers to reflect on a range of speclative security and privacy issues such as the portability of biometric identity, ambient surveillance, and the unintended consequences of personalization. Taken together, the collection demonstrates the breadth of concerns that can be explored through critical design in security and privacy contexts.

Second, this corpus serves as a proof-of-concept for our work on developing a structured approach to generating critical design probes for security and privacy. Rather than relying on purely ad hoc ideation, our process was guided by the systematic combination of four elements: (1) security and privacy concerns, misconceptions, or principles; (2) domains of application; (3) technological modalities or form factors; and (4) anticipated user tensions. By recombining these elements and iteratively refining resulting concepts, we produced a diverse set of artefacts that balance plausibility with critical tension. The resulting designs suggest that this combinatorial approach can support the generation of varied and conceptually distinct probes, offering a foundation for a more formalized methodology in future work.

2 Related Work and Research Positioning

[LEAH: TO DO]

Much has been written on critical design, and Malpass provides an excellent survey [11]. As our focus is security and privacy, we briefly summarize related concepts within and adjacent to critical design that address these issues. Critical design has long explored speculative artefacts as a means of provoking reflection rather than proposing deployable solutions. For example, Dunne and Gaver's *The Pillow* takes the form factor of a transparent pillow and plays radio signals it picks up from the technology of that time, such as phones, pagers, and baby monitors [3]; it challenges users to consider how confidential these signals are. Matsumoto's *Open Informant* is an app that "searches your communications for National Security Agency trigger words and then sends text fragments containing these words to the badge worn by the user for public display" [11]. Dunne and Raby's *United Micro-Kingdoms* reimagine the UK as four 'supershires' where one is occupied by 'digitarians:' extreme technocrats ruled by algorithms in a state of total surveillance.

While these designs demonstrate the value of critical artefacts in exposing hidden assumptions relating to security and privacy, we are unaware of any previous attempts to build a systematic set of designs central to this topic. However, critical design has been used in other areas of computer science. The uptake of critical design in HCI research is

encouraged based on the anticipated value critical design will add to the field [1]. Speculative design is another related concept that HCI researchers rely on. An example is to analyze and question the usage of a certain technology and the consequences of it in the near future [9]. Ludic design is also utilised to create designs that do not dictate a certain purpose or usage, but rather leave the interpretation to users [6]. Somewhat related to critical design, work on exploring alternative ways (like design fiction [10], role playing game [13], science fiction and design fiction [17]) to explain tricky concepts not only in security but also in other fields like sensing technologies. Studying the metaphors used in CS education gives important insights about how students perceive certain concepts [8].

Building on these traditions, this work explores how curated collections of critical design vignettes may function as communicative probes for reflecting on security and privacy risks. We structure this research as a two-phase design inquiry intended to move from exploratory artefact generation toward methodological formalization. The first phase focuses on creating and iteratively refining an initial corpus of speculative designs. The second phase (future work in §3.2) will develop a reproducible design procedure, articulating explicit steps and documentation practices to support adoption or adaptation by other researchers.

This paper reports on the first exploratory phase. Authors with cybersecurity research backgrounds collaboratively generated candidate artefacts through structured brainstorming informed by domain expertise. Ideation processes were captured on sticky notes on a virtual whiteboard and were organized into categories to scaffold ideation, including (1) security and privacy concerns (*e.g.,* 'if you don't pay for the product, you are the product'), misconceptions (*e.g.,* 'nothing to hide'), or principles (*e.g.,* 'minimal disclosure') that could be the basis of a critique; (2) domains where users might especially care about security or privacy (*e.g.,* health records or social networks); (3) a form factor for the new technology (*e.g.,* haptics or extended reality); and (4) a possible user response (*e.g.,* amused or embarrassed). After all ideas were captured and written as notes, they were randomly combined. We then drew notes at random (one note from each category) and held open-ended discussions about what the notes meant to us. Looking at these different combinations helped us brainstorm unconventional combinations to come up with compelling designs. We also individually brainstormed and shared our findings with the other authors in order to challenge the ideas we had developed. The combination of concepts was discussed in open-ended sessions to explore interpretive possibilities and identify promising narrative directions.During our open-ended discussions, when we could not come us with a design that meaningfully combined the notes we have drawn, we decided to focus on only one or two notes to ensure the designs met our criteria. Design concepts were then iteratively developed with the aim of producing near-future artefacts that remained plausible while foregrounding critical tension: *e.g.,* showing the darker side of convenience, risks that are normalized and overlooked, or ways a user might be betrayed by the technology. Concepts that failed to achieve conceptual clarity, plausibility, or distinctiveness were discarded. The resulting ten designs represent artefacts that survived multiple rounds of discussion and refinement, each requiring substantial time and revision. Either during the design of the artefact or after the vignette is finalized, we discussed potential titles that both represent the artefact and are engaging.. Next, we present our designs.

3 Initial Set of Designs

No(i)sy elevator. Elevator music has never been regarded as elevating your mood. This is why employees will be happier riding the noisy elevator. When it recognizes a user, it spins up a selection from their most-played songs on major music streaming platforms. Alice finds hearing her favourite Miles Davis song grounds her at the start of each workday, while Bob and Carol, coincidentally riding the elevator together, discover their shared love for 90s Britpop. David is a bit more concerned about the profanity-ridden banger that the nosy elevator plays for him and his boss.

Wifi Projector. Minimalist wifi routers have a single light indicator to show internet connectivity, such as a green light as opposed to red light. This understates the interesting and intriguing data flowing through the device as users browse the internet. By contrast, the wifi projector is a talkative router that projects visualizations onto the ceiling above where it is placed. Every website that pulls in scripts and cookies from other domains is displayed as a constellation of stars of various colours and sizes, which slowly fade away. While Alice's visit to dictionary.com is a beautiful galaxy, she is also perturbed by the number of ads and tracking companies surveilling her.

Crystal Avatars. Crystal avatars are created instantly but only get better over time. At first, they are not recognizable and cannot be differentiated from other users, but as users browse the site, small details crystallize. Alice is happy to save time registering for the site and have a personal avatar, which automatically adds a cat accessory after she spent a lunch hour reading about hairball treatments. However, she grows uneasy about the "Woman, Life, Freedom" badge, a cause she believes in but only vocalizes discriminately.

Chatty. The latest software update for Alice's voice-activated home assistant Chatty adds machine learning to better infer Alice's commands, such as adding items to her shopping list and suggesting music she might like. It does not always know whether Alice is talking to it or not, so over time it picks up on pieces of Alice's life and has trouble unlearning them. Alice is startled but pleased when Chatty chimes in with the name of 'that actor who was in that thing' she was telling her friend about. She is bemused to find a specialized vinyl cleaning solution on her shopping list after she said Bob's records smell fishy.

Dynamic Laughtrack. Digital television content, such as sitcoms, encode laughtracks as a series of cues rather than an actual recording. Smart TVs listen and classify the viewer's level of laughter on a 10-point scale (with fine-grained training over time). The TV adds laughter to the content in a sensitive and considerate manner, where the laughtrack is only marginally higher on the scale than the viewer's own laughter. This gently nudges the viewer to greater enjoyment of the program without bombarding her. It also mixes in actual past recordings of the viewer's own laughter to capitalize on emotional mirroring.

Receiver plant. The receiver plant in the front yard of Alice records the information of the people that pass by. It is a greedy plant that needs to absorb enough bluetooth data to grow. Bob is singlehandedly responsible for its lustrous canopy from walking by Alice's house every day, mostly while checking his work emails on his phone. Perhaps the plant overstepped when it displayed, "you are 15 minutes late today, are you sure you can make it to work on time?"

GOTTCHA. GOTTCHA is a human-detection system to protect online accounts being accessed by bots. It invokes the device camera to analyze if a live human is using the device. To protect against video replays or machine-generated video, it unexpectedly prompts the user with a randomly selected image and carefully measures their reaction to it. Micro-expressions are involuntary facial displays of emotion that are too fast to mimic artificially. GOTTCHA's image bank can provoke disgust, anger, fear, sadness, happiness, surprise or contempt. Over time, Alice stops visiting websites that use GOTTCHA because of its capricious tendency to mix in disturbing images.

ToS Fishing. ToS Fishing is a digital game where players earn points by hunting down excessive legalese on websites, including terms of service (ToS), privacy policies, and cookie policies. To play, users simply copy and paste the URL of the legalese. If the ToS has been seen before, the users are awarded points based on its word count. If it has not been seen before, it is reviewed by a human for validity and word count—and in this case, the user gets a finder's bonus. To mitigate people from launching custom websites, the game only accepts sites from the Alexis top million. Users display their aggregate score on a leaderboard with a profile showing their top catches (word count only, not the website), and can earn various badges for playing consistently.

Eleventh Finger. Biometric authentication based on fingerprints is generally user-friendly and fast. Eleventh finger is a 3D printed rubber finger, customized with the user's fingerprint. You can put it on a keychain and use it on cold winter days when you do not want to remove your gloves. Alice lets her friend Bob borrow it when he stays at her house. It can also serve as a backup if the worst happens.

Calm Watch. Calm Watch is a smartwatch that tracks anxiety levels based on several biomarkers. When the anxiety level of the wearer crosses a threshold, a haptic vibration pulse prompts the user to look at the watch. From there, they can select a variety of calming techniques, including breathing exercises and guided meditation. Bob is nervous to talk to Alice and becomes flustered when his watch starts vibrating mid-conversation, just loud enough for Alice to hear it and notice his fidgeting with the watch.

3.1 Discussion

Vignettes. Critical designs are commonly described through vignettes, though they are occasionally built and displayed as physical artifacts. For example, Dunne and Raby's "Compass Table" in Table 1 (and the seven other objects in their *Placebo* project [4]), were built and given to individuals to use, followed by exit surveys capturing how the people felt about the objects. We presented our designs as short vignettes that describe both their form and use context. When successful, such descriptions allow readers to readily imagine these artefacts within their own lives and engage with them critically, resulting in their 'rhetorical use' [11] rather than practical deployment. These designs are intentionally transgressive, crafted to provoke emotional responses by invoking humour, satire, ridicule, poetry, playfulness, lewdness, appropriation, irreverence, or deliberate overcompensation.

Vignettes used in research contexts, in comparison, are defined as "short stories about hypothetical characters in specified circumstances, to whose situation the interviewee is invited to respond" [5] are established instruments in speculative design, design fiction, and scenario-based inquiry in HCI. In these contexts, narrative depictions of hypothetical technologies or practices are used to surface assumptions, value tensions, and reflections on possible futures without requiring the artefact itself to exist materially. Design fiction, in particular, is not treated as empirical stimuli for behavioural measurement, but as structured prompts that enable participants to reason about situated technological

futures, their consequences, and social norms. Vignette-based methods in usable security research (e.g., [14], [7]) similarly construct narratives to investigate user expectations, decision-making processes, and ethical considerations surrounding emerging technologies when direct observation or deployment is infeasible, unsafe, or premature.

Usage Contexts. In this work, we propose vignettes as a methodological vehicle for communicating speculative artefacts rather than as experimental instruments intended to elicit measurable responses. The vignettes function as shareable analytic artefacts that others may appropriate for reflection, inquiry, and design exploration. This approach therefore aligns with established HCI practices in which scenarios are used to probe socio-technical implications.

Beyond their role as speculative artefacts, we foresee several practical use cases of critical design vignettes for security and privacy. First, they may serve as reflective prompts in educational, participatory, or professional contexts, where speculative scenarios facilitate discussion of assumptions, values, and mental models surrounding security and privacy practices. Second, they may function as research instruments in future speculative design or vignette-based studies, where they can be adapted as stimuli to explore perceptions of risk, trust, or technological futures. Finally, we envision their use as generative tools in design processes to inspire ideation, highlight overlooked attack vectors, or prompt reconsideration of design assumptions before technologies are implemented.

We therefore position 'Security Noir' (an homage to Dunne and Raby's term 'Design Noir' [4]) as a critical design approach for engaging with the often uncomfortable or hidden aspects of security and privacy technologies and their impacts on human experience. Methodologically, this approach takes the form of a lightweight, imaginable, and scalable medium that allows speculative artefacts to be encountered without technological implementation, while remaining grounded in established HCI traditions of using vignettes in survey research, design fiction, and scenario-based studies. These critical designs intend to activate readers' existing mental models and experiences, supporting reflective engagement with security and privacy risks that are otherwise difficult to visualize or experience directly.

3.2 Future Work

As a work-in-progress, this paper reports on the initial exploratory phase of our approach. Future work will focus on formalizing the design process into a reproducible method, expanding the space of concerns and domains considered, and evaluating how such vignettes function in practice (e.g., in workshops, educational settings, or participatory design contexts). Through this progression, we aim to establish critical design vignettes as a viable methodological tool within usable security and HCI research.

First, while this work demonstrates our approach based on concerns, domains, modalities, and user tensions, future work will expand and refine these dimensions through a more comprehensive synthesis of usable security and privacy literature. This includes developing more exhaustive and theoretically grounded sets of security and privacy concerns, misconceptions, and principles, as well as broadening the range of stakeholders and contexts.

Second, we plan to more systematically document and analyze the design process itself. In this work, ideation and refinement were conducted through iterative discussion and selection; future work will more explicitly capture this process through structured design tools (e.g., ideation cards or generative frameworks) and reflective documentation. This will enable us to better articulate how concepts are generated, filtered, and developed, addressing current limitations in methodological transparency and supporting adoption by other researchers. For example, we have begin to develop structured ideation cards representing concepts, misconceptions, domains, and interaction modalities. These cards will be combined in small sets to scaffold exploration and examine how constrained prompts shape design diversity, thematic coverage, and narrative plausibility. We will also explore generative AI as an ideation partner (the designs

in this paper were developed without AI assistance). To better understand these processes, we plan to document the design work through autoethnography to identify recurring practices and challenges, including topic selection and balancing plausibility with provocation.

Third, we aim to evaluate how these vignettes function in practice in user studies. While this paper positions them as communicative probes, our future work will examine how different audiences interpret and engage with the designs, such software develpers and other non-specialists. In particular, we are interested in how vignettes shape users' threat identification of security and privacy risks, what kinds of discussions they provoke, and whether they reveal gaps, misconceptions, or alternative perspectives that are not easily captured through traditional methods.

Finally, we will explore how critical design vignettes can be integrated into existing security practices, including threat modelling and design ideation workflows. Given the inherently speculative nature of threat identification, we see potential for these artefacts to complement conventional approaches by supporting early-stage exploration of "what can go wrong", as suggested by Shostack + Associates [16] [CITE Shostack 2021], in ways that are accessible to non-specialists. Future work will investigate how such methods can be incorporated into interdisciplinary teams and whether they can meaningfully inform design and decision-making processes. Through these directions, we aim to move from an initial corpus of designs toward a more robust methodological framework for using speculative artefacts as tools for reflection, communication, and inquiry within usable security and HCI research.

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